

eMargin: Revisiting Contrastive Learning with Margin-Based Separation

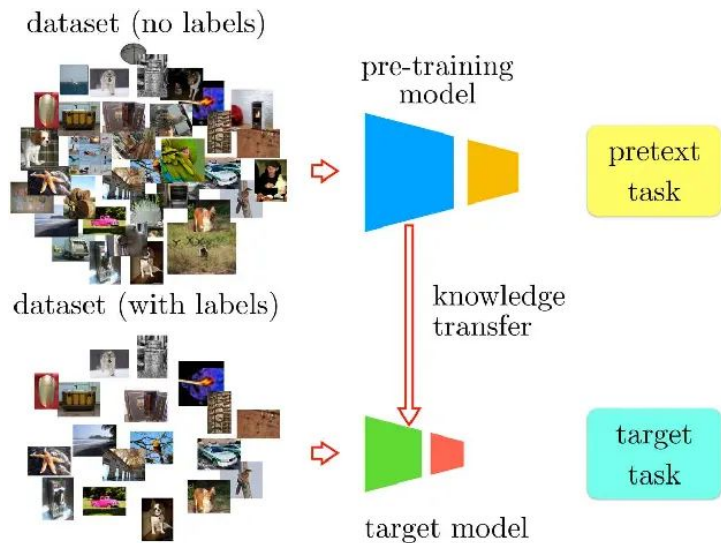
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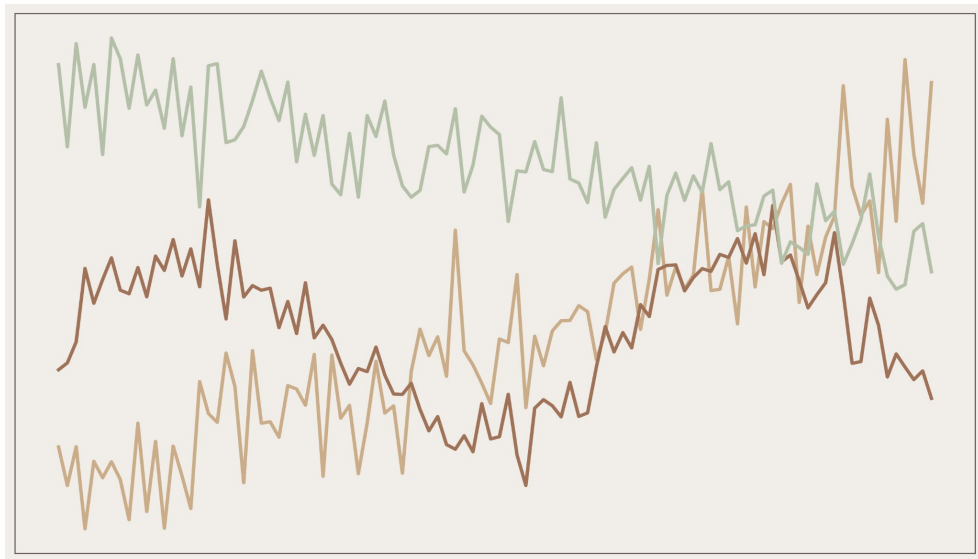
Learning from Difficult Data Workshop (LDD at ECAI 2025)

Self-supervised learning motivation



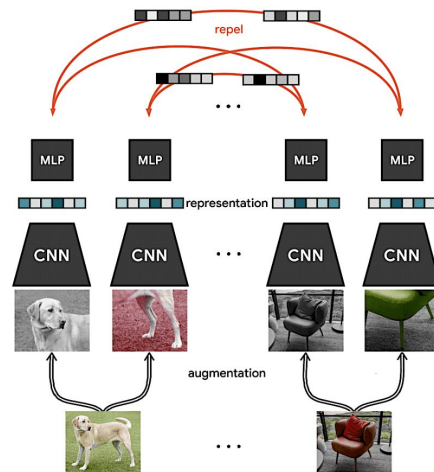
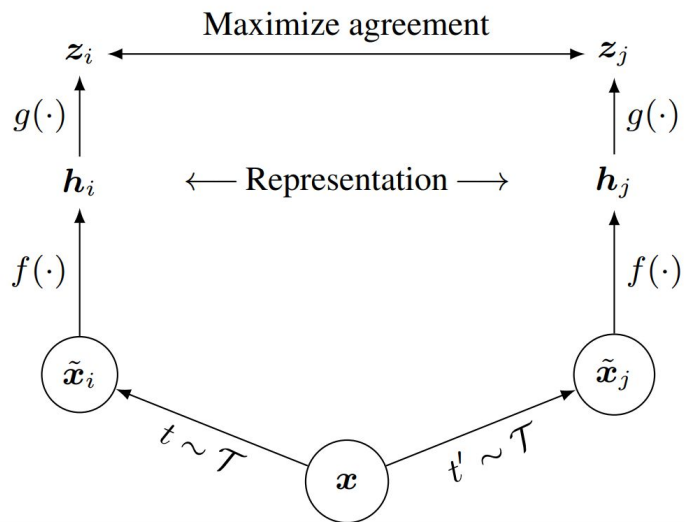
- Supervised learning
 - Relies on large labeled datasets
 - Expensive and domain-specific annotations
 - Limited generalization

Self-supervised learning motivation



- Supervised learning
 - Relies on large labeled datasets
 - Expensive and domain-specific annotations
 - Limited generalization
- Self-supervised learning
 - Learns representations directly from data
 - Scalable (unlabeled data is abundant)
 - Improves transferability to downstream tasks

Contrastive representation learning



Contrastive learning in time series

TNC, MF-CLR, CoST, InfoTS, TS2Vec

- Augmentations and masking
- Dynamic Time Warping (DTW)
- Multiple models to capture granularity

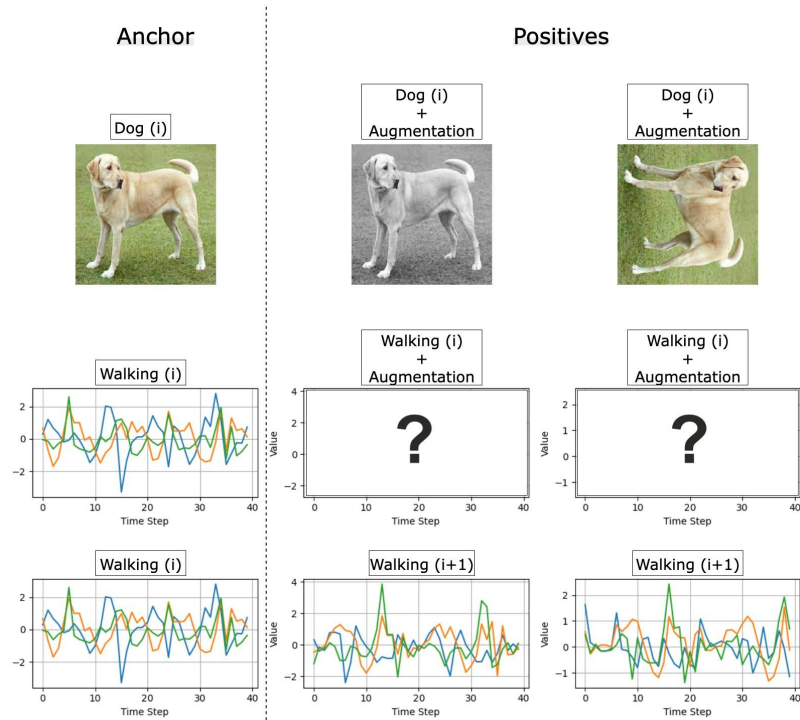
Consequence

- Data distortion
- Scalability and speed

Looking at the temporal information

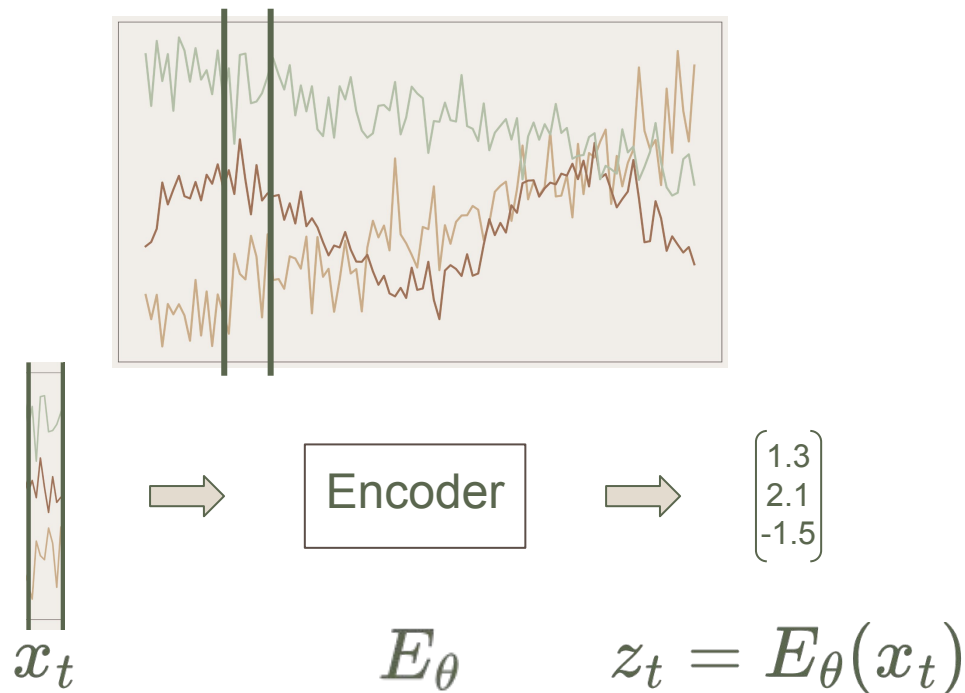
Similarity within nearby instances

- That consecutive instances in a sequence have the same class and event labels would not change too often
- Often holds for time series, which have repeated labels in the temporal dimension



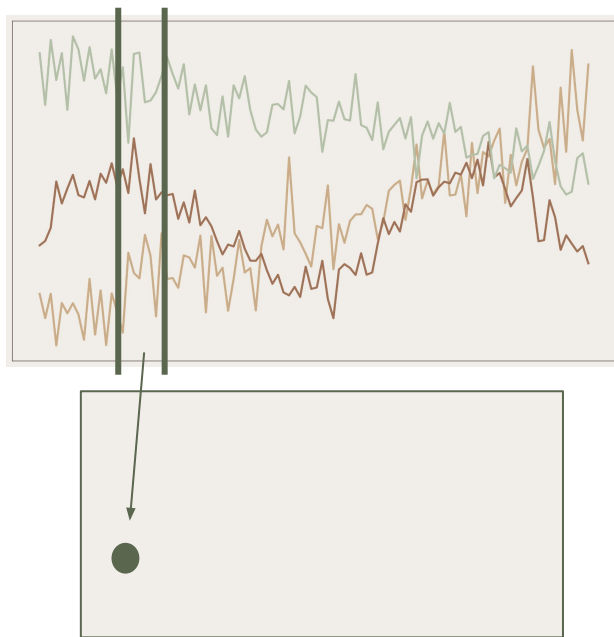
Looking at the temporal information

FIRST: Learn an embedding for small segments (instances) of time series in an unsupervised way so things that appear similar are close in some space



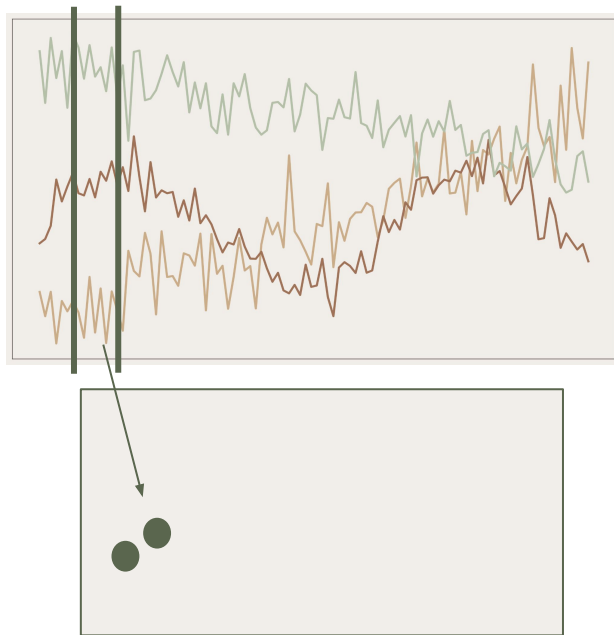
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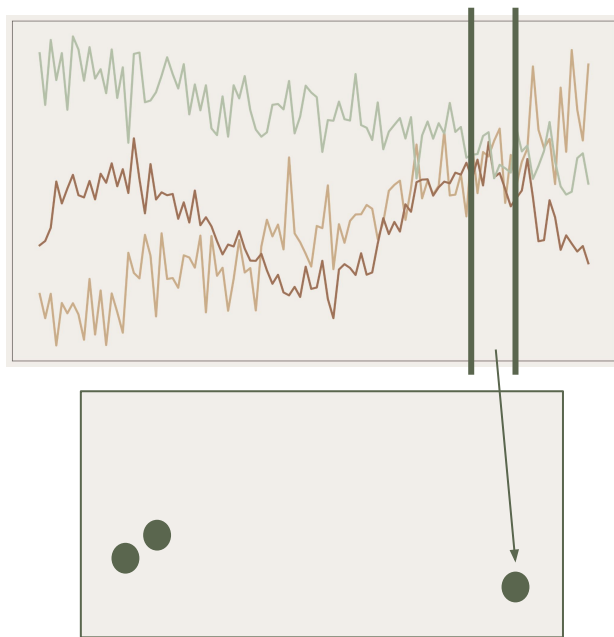
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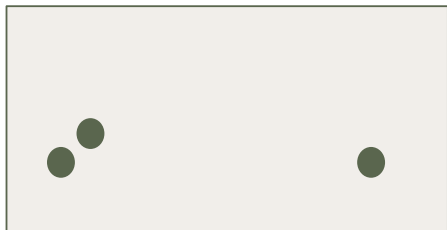
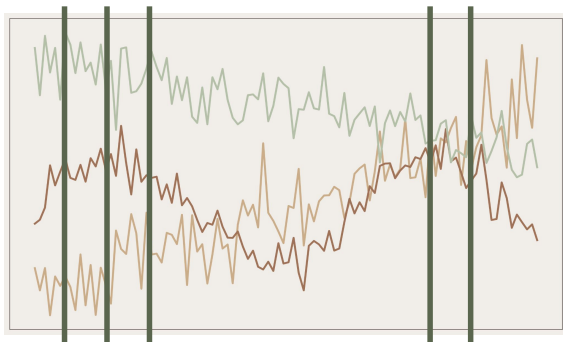
Looking at the temporal information

FIRST: Learn an embedding for small segments (instances) of time series in an unsupervised way so events that appear similar are close in the embedding space



Looking at the temporal information

- Discard Augmentation
- Pattern within the data (Similarity within nearby instances)
- Scalable ?



Contrast everything!

What happens at transition?
(eMargin)

Contrastive Learning

$$L_{\text{InfoNCE}} = -\mathbb{E} \left[\log \frac{\exp(\text{sim}(z_t, z_{t+1})/\tau)}{\sum_{k \neq t, t+1} \exp(\text{sim}(z_t, z_k)/\tau)} \right]$$

where:

- z_t and z_{t+1} are positive pairs (e.g., adjacent time steps),
- z_k are negative samples,
- $\text{sim}(z_t, z_{t+1})$ is the cosine similarity,
- τ is the temperature parameter,

Margin in Contrastive Learning

We define a similarity matrix $\mathbf{M} \in \mathbb{R}^{T \times T}$, where element $\mathbf{M}_{t,t+1}$ stores the similarity score between consecutive time steps:

$$\mathbf{M}_{t,t+1} = \text{sim}(z_t, z_{t+1})$$

To prevent representation collapse while ensuring sufficient separation for dissimilar samples, we modify this matrix using an adaptive margin penalty:

$$\mathbf{M}_{\text{margin}} = \frac{1}{2}(1 - \mathcal{Y})\mathbf{M}^2 + \frac{1}{2}\mathcal{Y} [\max(0, \text{margin} - \mathbf{M})]^2 \quad (2)$$

Where, $\mathcal{Y} = \begin{cases} 0, & \text{if } \text{sim}(x_t, x_{t+1}) > \text{threshold} \\ 1, & \text{otherwise} \end{cases}$

Margin in Contrastive Learning

- When $\text{sim}(x_t, x_{t+1})$ is high ($\mathcal{Y} = 0$):

$$\mathbf{M}_{\text{margin}} = \frac{1}{2} \mathbf{M}^2$$

This maintains the similarity structure for strongly correlated time steps, reinforcing local continuity.

- When $\text{sim}(x_t, x_{t+1})$ is low ($\mathcal{Y} = 1$):

$$\mathbf{M}_{\text{margin}} = \frac{1}{2} [\max(0, \text{margin} - \mathbf{M})]^2$$

Margin in Contrastive Learning

for each consecutive pair (z_t, z_{t+1}) in sequence **do**

 Apply margin adjustment:

$$\mathbf{M}_{\text{margin},t,t+1} \leftarrow \begin{cases} \frac{1}{2} \mathbf{M}_{t,t+1}^2, & \text{if } \mathcal{Y} = 0 \\ \frac{1}{2} [\max(0, \text{margin} - \mathbf{M}_{t,t+1})]^2, & \text{if } \mathcal{Y} = 1 \end{cases}$$

 Compute InfoNCE + eMargin:

$$L_{\text{eMargin},t,t+1} = -\mathbb{E} \left[\log \frac{\exp(\mathbf{M}_{\text{margin},t,t+1}/\tau)}{\sum_{k \neq t,t+1} \exp(\mathbf{M}_{\text{margin},t,k}/\tau)} \right]$$

end for

Downstream tasks - Classification

ECG

Normal heartbeat



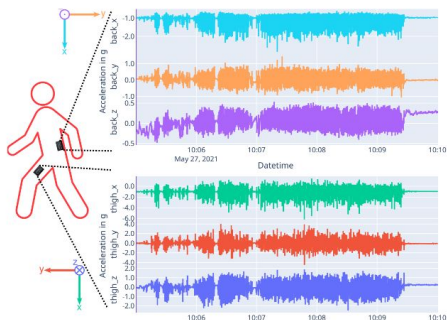
Atrial flutter



Atrial fibrillation

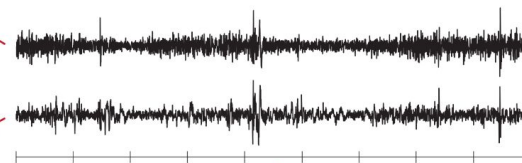
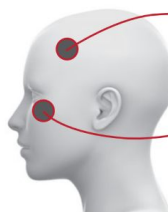


HARTH

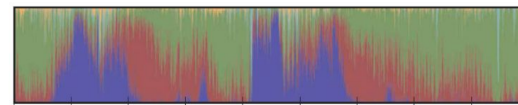


	# Instance	Sequence length	Dimension	Classes
HARTH	1,270,087	119	156	12
SLEEP	478,600	200	90	5
EEG	255,200	200	1502	4

SleepEEG



High-Frequency Sleep Stage Representation
128 Hz



Segment Classification
E.g. 1/30 Hz



Clusterability

Davies-Bouldin Index (DBI) and Silhouette Score

Table 1: Clustering performance across three datasets (HARTH, SleepEeg, and ECG) is evaluated using the Davies-Bouldin Index (DBI, lower is better) and Silhouette score (higher is better).

	HARTH		SLEEPEEG		ECG	
	DBI↓	Silhouette ↑	DBI↓	Silhouette ↑	DBI↓	Silhouette↑
TNC	1.10±0.11	0.40±0.08	1.18±0.05	0.41±0.10	1.06±0.13	0.44±0.05
InfoTS	1.17±0.07	0.29±0.00	0.85±0.04	0.48±0.01	0.73±0.18	0.56±0.03
CoST	1.91±0.12	0.24±0.06	1.21±0.03	0.40±0.01	1.61±0.25	0.33±0.02
SimMTM	1.98±0.11	0.75±0.05	1.77±0.19	0.49±0.05	2.42±0.15	0.17±0.01
TimeDRL	1.46±0.12	0.30±0.04	0.65±0.03	0.62±0.09	0.96±0.11	0.45±0.05
TS2Vec	1.17±0.06	0.33±0.02	1.11±0.05	0.33±0.00	0.94±0.03	0.45±0.02
SoftCLT	1.37±0.01	0.28±0.02	1.46±0.10	0.24±0.02	1.06±0.07	0.39±0.02
MF-CLR	1.81±0.10	0.23±0.02	1.10±0.12	0.43±0.02	1.27±0.14	0.40±0.02
InfoNCE + <i>eMargin</i>	0.97±0.07	0.38±0.01	0.62±0.05	0.83±0.04	0.59±0.02	0.80±0.01

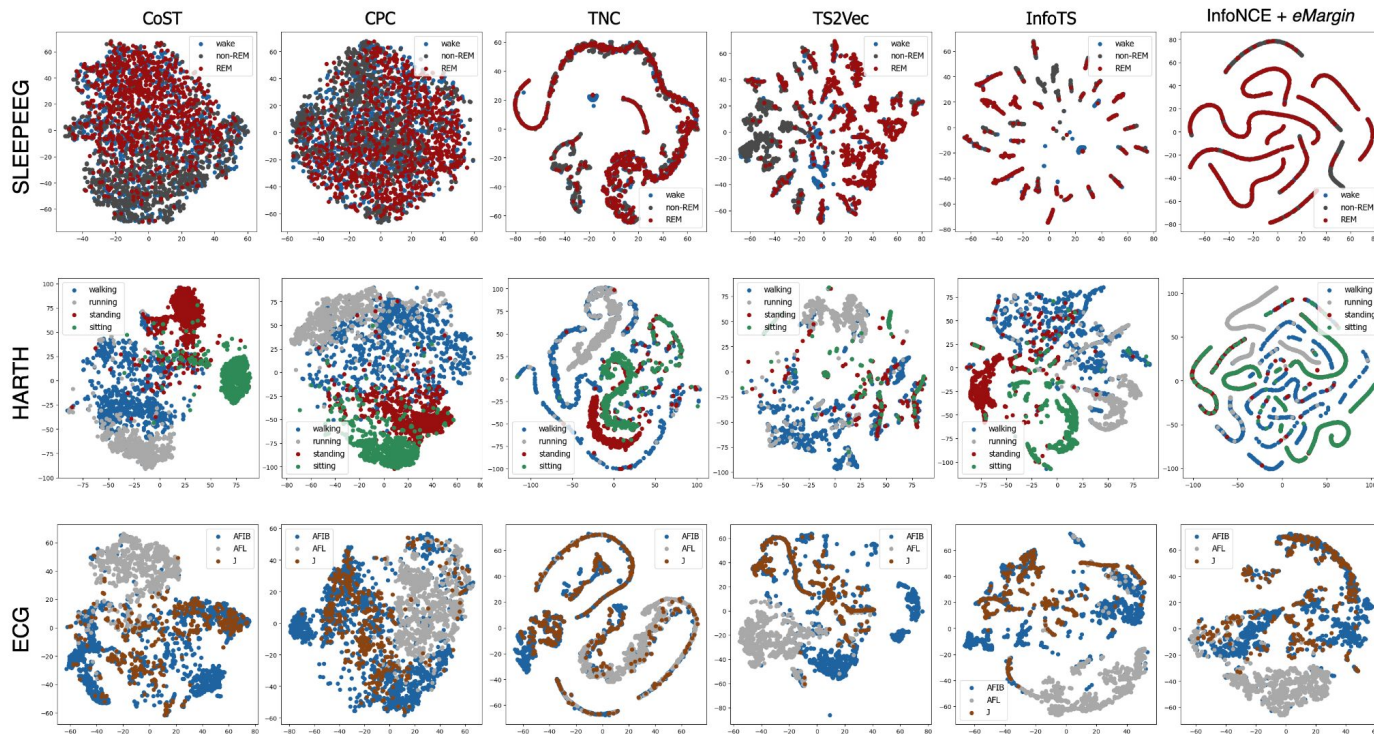
Linear evaluation with frozen backbone

Table 2: Comparison with state-of-the-art methods on linear evaluation with a frozen backbone. We compare eMargin with baselines and a randomly initialized encoder (*Random Init.*) on frozen evaluation. We train a linear classifier on top of the features from the encoder using the entire train set and evaluate the test set.

Datasets	Models	Accuracy	F1 score	Precision	Recall
HARTH	Random Init.	86.69±0.50	0.84±0.01	0.55±0.01	0.45±0.02
	TNC	89.03±0.03	0.88±0.00	0.70±0.02	0.55±0.01
	InfoTS	90.12±0.13	0.89±0.00	0.71±0.02	0.56±0.01
	CoST	88.68±0.41	0.88±0.00	0.64±0.01	0.56±0.00
	SimMTM	68.14±4.83	0.60±0.08	0.34±0.04	0.24±0.05
	TimeDRL	71.92±2.37	0.63±0.04	0.37±0.04	0.28±0.01
	TS2Vec	87.59±0.23	0.85±0.00	0.58±0.01	0.49±0.00
	TS2Vec + <i>SoftCLT</i>	88.07±0.13	0.86±0.00	0.62±0.03	0.50±0.01
	MF-CLR	77.07±5.38	0.73±0.07	0.52±0.04	0.37±0.05
	InfoNCE + <i>eMargin</i>	87.87±0.54	0.86±0.01	0.56±0.04	0.48±0.03

Visualization of learned representations

t-SNE plots of the learned embeddings



Findings

- Our results highlight critical discrepancies between unsupervised metrics, representation structure, and downstream task performance
- Optimization objective may prioritize geometric compactness and separation without capturing the underlying semantic structure necessary for generalization

Achieving high scores on unsupervised clustering metrics does not necessarily imply that the learned embeddings are meaningful

Paper:

<https://arxiv.org/abs/2507.14828>



Code:

<https://github.com/sfi-norwai/eMargin>

